

针对8天循环训练计划 V6 版(四维目标:增肌、心肺/VO2 Max、爆发力/防跌倒、骨密度/平衡)的核心要求,对代码项目进行全面的系统重构。

一、V6 升级架构与需求映射



1. 绿灯 / 黄灯 / 红灯状态机逻辑

- 绿灯模式(满血 / 标准状态) :
 - 开放自由重量与器械等效选项(并列任选)。
 - 首个复合动作正式组前 2 次自动激活「最大意图向心爆发(~1s 向心, 2-3s 离心控制, 第 3 次起匀速)」提示与节拍辅助。
 - 自由动作提示做 1 组约 15 次的轻重量过渡组, 执行满额容量与 RPE 8 双区间。
- 黄灯模式(疲劳 / 状态欠佳 / 关节轻微不适):
 - 自动退守纯器械版, 隐藏或禁用自由重量切换;
 - 自动阻断爆发意图, 全程匀速控制, 避免动作变形;
 - 建议容量自动下调 1-2 组(或直接截断尾组), 保留次数提升至 RPE 7-8。
- 红灯模式(极度疲劳 / 生病 / 关节急性疼痛):
 - 主课表自动折叠, 替换为「防衰主动恢复卡」(20-30分钟低心率轻松散步、轻柔动态拉伸、泡沫轴筋膜放松或 20 分钟太极);
 - 记录当日休整状态, 保障系统打卡连续性, 阻断负重损伤。

2. 滑动备忘卡区(Memo Deck Carousel)

- 顶部提供常驻快捷胶囊按钮  V6 备忘智库, 点击滑出全屏/底部卡片抽屉。
- 支持横向平滑滑动(Touch Scroll Snap / 左右切卡), 分为 5 大卡片分类:
 1. 器械等效替代卡: D1/D2/D4/D5 自由动作与器械并列对应关系、过渡组规则。
 2. Linda 死手模式边界卡: 金属宽面钩适用范围、开放 RDL 与六角杠、严禁动作清单、非训

练期独立握力器补偿方案。

3. 骨密度与冲击训练卡:大肌群复合加重、D3 坡度快走、提踵重要性、跳跃冲击进阶阶梯(脚跟落地 → 台阶上下 → 原地小跳)。
4. 平衡与功能指标卡:热身 5 分钟单腿站立/足跟接足尖步态、年度握力阈值(男 <26kg, 女 <16kg)、季度单腿站立睁眼测试(目标 >30s)。
5. 四条不可压缩铁律卡:D6至下轮D1满72小时、有氧前一天非腿日、完整平衡热身、4×4每循环必做。

3. V6 动作与规则深度调整

- 时长校准:力量日放宽至 71-73 分钟, D3 Zone 2 调整为 55 分钟(含核心与10分钟拉伸), D6 4×4 调整为 40 分钟。
- 热身结构重构:由原常规热身调整为 12 分钟, 末尾 5 分钟强制固化为「单腿站立 3×30s/侧 + 足跟接足尖行走 2×10步 + 侧向跨步 2×10步」的平衡与神经肌肉打卡项。
- 休息日日常化:D7、D8 休息日新增「20-30分钟日常轻松散步(心率<100)」打卡选项, 补足中等强度有氧频率。

二、核心数据模型配置 (v6-workout-data.js)

将原本硬编码或器械优先的数据结构重构为符合 V6 规范的数据定义:

```
JavaScript
// v6-workout-data.js
export const V6_PLAN_CONFIG = {
  version: "V6.0",
  cycleDays: 8,
  title: "8天循环训练计划 V6(增肌+防衰四维双目标版)",
  dateRange: "2026-10-01 至 2026-11-25",
  nonCompressibleRules: [
    "D6 到下一轮 D1 必须间隔满 72 小时",
    "Zone 2 与 4×4 的前一天绝对不能是腿日",
    "首个复合主项前必须完成含 5 分钟平衡与步态模块的完整热身",
    "4×4 每循环必须做满 1 次, 减载轮亦不例外"
  ],
}

// 备忘智库卡片数据
export const MEMO_CARDS = [
  {
    id: "memo-swap",
    tag: "器械等效",
    title: "自由动作 ≡ 器械等效替代(绿灯日并列)",
    color: "blue"
  }
]
```

```
content
  day "D1 下肢A" machine "哈克深蹲机" free "史密斯机深蹲 / 六角杠硬拉" leo true linda true
  day "D2 上肢A" machine "挂片推胸机" free "杠铃/哑铃卧推(Linda禁用)" leo true linda false
  day "D2 上肢A" machine "坐姿推肩机" free "站姿哑铃/杠铃推举(Linda禁用)" leo true linda false
  day "D4 下肢B" machine "臀推机 B-Stance" free "杠铃 RDL / 六角杠 RDL(最友好选项)" leo true linda true
  day "D5 上肢B" machine "高位下拉机" free "助力引体 / 自体重引体(Linda禁用)" leo true linda false
  day "D5 上肢B" machine "挂片划船机" free "哑铃单臂划船 / 杠铃俯身划船(Linda禁用)" leo true linda false
  day "强化项" machine "(无对应)" free "农夫行走(强烈推荐 Leo; Linda禁用)" leo true linda false
rules
  "绿灯日自由与器械完全平等, 任选其一; 黄灯日强制退守器械版。"
  "选择自由版动作前, 必须额外增加 1 组约 15 次的轻重量过渡组。"
  "自由动作严守 RPE 8 与双区间铁律, 离心严格保持 2-3 秒控制。"
id "memo-linda"
tag "死手守则"
title "Linda 死手模式(Grip-Free)边界指南"
color "rose"
badge "避免小臂变粗 · 阻断斜方代偿"
content
  "【已开放自由动作】: 杠铃 RDL、六角杠硬拉、史密斯深蹲(需全程使用金属宽面钩助力连接杆)。"
  "【六角杠RDL优势】: 躯干比直杠更直立, 腰椎剪切力显著降低, 为 Linda 最优自由髋主导选项。"
  "【绝对禁用清单】: 哑铃侧平举、哑铃划船、自由卧推、农夫行走(禁止任何需握紧完成的动作)。"
  "【独立握力代偿方案】: 非训练日独立使用轻负荷握力器(3组 × 15-20次), 不介入主项动作链。"
id "memo-power-bone"
tag "防衰爆发"
title "爆发力向心 & 骨密度推进指南"
color "amber"
content
  "【爆发力意图】: 每个力量日首个复合动作正式组前 2 次, 以『最大意图最快速度(~1s)』向心推起/蹬起, 离心仍保持 2-3s 控制; 第 3 次恢复常规节奏。零额外加组与时间成本。"
  "【骨密度动作库】: 大肌群复合加重(哈克/倒蹬/RDL/推胸/划船) + D4/D5 提踵(保留刺激髋、脊柱、
```

踝)。”

“【负重有氧原则】:D3 采用坡度快走(绝经后女性增骨密度 0-2%)。椭圆机为下肢减负设计, 骨密度收益低于快走。”

“【冲击跳跃阶梯(无骨松且医嘱许可)】:脚跟落地 2×10 步 → 台阶上下 2×20 步 → 原地小跳 2×10 次(稳定4-6周方可进阶)。”

id "memo-metrics"

tag "基线测试"

title "功能性指标与筛查备忘"

color "emerald"

content

“【握力年度检测】:每年测定 1 次。肌力低下警示阈值:男性 < 26kg, 女性 < 16kg。”

“【单腿站立睁眼测试】:每季度测试 1 次。闭眼/睁眼单腿站立目标 > 30 秒(防跌倒前庭本体觉)。”

“【DXA 骨密度检测】:每 1-2 年复查腰椎与髋部 T 值基线, 确认抗阻负荷进度保守程度。”

“【心血管与代谢】:4×4 前需完成心血管风险筛查;日常监测血压血脂, 排查打鼾睡眠呼吸暂停。”

id "memo-rules"

tag "不可压缩"

title "8天循环四项不可压缩铁律"

color "purple"

content

"1. D6(4×4)到下一轮 D1(下肢A)之间必须满打满算休足 72 小时。"

"2. Zone 2 与 4×4 高强度有氧的前一天, 绝对不能安排腿部力量训练。"

"3. 力量日首个复合主项前, 必须执行包含 5 分钟平衡与步态的完整热身。"

"4. 4×4 HIIT 每循环必须执行满 1 次, 即使进入减载周也不得剔除。"

// 8天训练课表 V6 详细架构

export const V6_DAYS_SCHEDULE =

day "D1"

name "下肢 A(股四主导 + 臀)"

type "strength"

duration "73 分钟"

warmup

dynamicMin 7

balanceMin 5

```
balanceltems
"单腿站立(可微扶墙, 渐进脱手):3组 × 30秒/侧"
"足跟接足尖直线脚跟走(Tandem Gait):2组 × 10步"
"侧向重心跨步控制:2组 × 10步"

exercises
id "d1-ex1"
name "哈克深蹲机"
isFirstCompound true // 触发爆发力意图
freeEquiv "史密斯机深蹲 / 六角杠硬拉"
sets 4
reps "8-10"
rpe 8
lindaAllowedFree true
rest "2-3 分钟"
notes "首动作:正式组前2次执行最快向心意图(~1s), 离心控制2-3s。若选自由版需加做1组15次轻重量过渡组。"

id "d1-ex2"
name "器械腿屈伸"
isFirstCompound false
sets 3
reps "10-12"
rpe 8
rest "90 秒"
notes "顶峰收缩 1 秒, 专注股四头肌离心受控。"

id "d1-ex3"
name "45度倒蹬机(双脚高位偏宽)"
isFirstCompound false
sets 3
reps "10-12"
rpe 8
rest "2 分钟"
notes "主打臀大肌与股四头肌协同, 骨密度重要承重项。"

id "d1-ex4"
name "器械髋外展(外扩)"
```

```
isFirstCompound false
sets 3
reps "12-15"
rpe 8
rest "60 秒"
notes "骨盆稳定肌与臀中肌强化, 防跌倒重要基石。"
cooldown "3 分钟股四与臀大肌简短静态拉伸"
day "D2"
name "上肢 A(推优先 + 背宽 )"
type "strength"
duration "72 分钟"
warmup
dynamicMin 7
balanceMin 5
balanceItems "单腿平衡站立:3 × 30s/侧" "足跟接足尖走:2 × 10步"
exercises
id "d2-ex1"
name "挂片推胸机"
isFirstCompound true
freeEquiv "平凳杠铃卧推 / 哑铃卧推(Linda禁用)"
sets 4
reps "8-10"
rpe 8
lindaAllowedFree false // Linda死手模式禁用
rest "2 分钟"
notes "正式组前2次最大意图向心爆发推起。Linda统一使用挂片推胸机。"
id "d2-ex2"
name "坐姿推肩机"
isFirstCompound false
freeEquiv "站姿杠铃/哑铃推举(Linda禁用)"
sets 3
reps "10-12"
rpe 8
lindaAllowedFree false
rest "90 秒"
```

```
notes "前三角肌与上胸主导, 保护肩峰撞击。"
id "d2-ex3"
name "宽距高位下拉机"
isFirstCompound false
freeEquiv "助力引体向上"
sets 4
reps "8-10"
rpe 8
lindaAllowedFree false
rest "2 分钟"
notes "背阔肌宽度主导, 沉肩下压拉至上胸。"
id "d2-ex4"
name "绳索侧平举"
isFirstCompound false
sets 3
reps "12-15"
rpe 8
rest "60 秒"
notes "中束孤立, 全程离心对抗。"
cooldown "3 分钟胸大肌与背阔肌拉伸"
day "D3"
name "Zone 2 低强心肺 + 核心拉伸"
type "cardio"
duration "55 分钟"
warmup dynamicMin 5 balanceMin 0 balancetems
cardioProtocol
target "心率维持在 Zone 2 区间(建议 110-125 bpm, 依个体换算)"
main "跑步机 6-9% 坡度快走 40 分钟(促骨密度, 绝经期与中老年优于椭圆机)"
core "死虫式 3×12次 + 侧平板支撑 3×30秒"
stretch "全身系统性柔韧拉伸 10 分钟"
day "D4"
name "下肢 B(后链腘绳肌/臀 + 手臂)"
```

```
type "strength"
duration "73 分钟"
warmup
dynamicMin 7
balanceMin 5
balanceItems
"单腿站立睁眼测试/练习:3组 × 30秒/侧"
"足跟步态练习:2组 × 10步"
"动态侧向跨步重心转移:2组 × 10步"
exercises
id "d4-ex1"
name "臀推机 B-Stance(单腿主导)"
isFirstCompound true
freeEquiv "杠铃 RDL / 六角杠 RDL(★Linda已纳入, 金属钩可用)"
sets 4
reps "8-10"
rpe 8
lindaAllowedFree true // Linda 绿灯开放六角杠或直杠RDL
rest "2 分钟"
notes "首动作爆发意图!六角杠RDL为腰椎最友好自由后链选项。强化脊柱抗屈曲能力。"
id "d4-ex2"
name "俯卧/坐姿器械腿弯举"
isFirstCompound false
sets 4
reps "10-12"
rpe 8
rest "90 秒"
notes "腓绳肌孤立, 慢速离心下落。"
id "d4-ex3"
name "站姿提踵(台阶或器械)"
isFirstCompound false
sets 4
reps "12-15"
rpe 8
rest "60 秒"
notes "ACSM 明确骨密度与踝关节防扭伤关键动作, 严禁砍掉。"
```

```

[
  {
    id "d4-ex4"
    name "绳索三头下压 + 弯举"
    isFirstCompound false
    sets 3
    reps "12-15"
    rpe 8
    rest "60 秒"
    notes "手臂功能维持与肘关节韧带保护。"
  },
  {
    cooldown "3 分钟腓绳肌与小腿腓肠肌拉伸"
  },
  {
    day "D5"
    name "上肢 B(拉优先 + 胸厚度 + 三角肌后束)"
    type "strength"
    duration "71 分钟"
    warmup
    dynamicMin 7
    balanceMin 5
    balanceItems "单腿站立平衡:3 × 30s/侧" "脚尖接脚跟行走:2 × 10步"
  },
  {
    exercises
    {
      id "d5-ex1"
      name "胸承托挂片划船机"
      isFirstCompound true
      freeEquiv "哑铃单臂划船 / 杠铃俯身划船(Linda禁用)"
      sets 4
      reps "8-10"
      rpe 8
      lindaAllowedFree false
      rest "2 分钟"
      notes "首复合动作向心爆发意图拉起。胸托阻断下背代偿。"
    },
    {
      id "d5-ex2"
      name "蝴蝶机夹胸 / 上斜器械推胸"
      isFirstCompound false
      sets 3
      reps "10-12"
    }
  }
]
```

```
rpe 8
rest "90 秒"
notes "胸大肌厚度补充。"
id "d5-ex3"
name "反向蝴蝶机后束飞鸟"
isFirstCompound false
sets 3
reps "12-15"
rpe 8
rest "60 秒"
notes "圆肩驼背矫正与后肩袖强化。"
id "d5-ex4"
name "倒蹬机小腿提踵(补充)"
isFirstCompound false
sets 3
reps "15-20"
rpe 8
rest "60 秒"
notes "下肢骨密度与跟腱刚度补充强化。"
cooldown "3 分钟上肢综合拉伸"
day "D6"
name "4×4 挪威高强度间歇(HIIT)"
type "hiit"
duration "40 分钟"
warmup dynamicMin 8 balanceMin 0 balancetems
cardioProtocol
target "冲刺段心率达到 85-95% HRmax(强力提升 VO2 Max)"
protocol "4 轮 ×(4分钟高强度冲刺 + 3分钟低强度活动恢复)"
cooldown "5 分钟慢走 + 5 分钟下肢系统拉伸"
rule "执行后到下一循环 D1 必须满 72 小时物理恢复!"
day "D7"
name "主动休整日 A"
```

```

    type "rest"
    duration "自由安排"
    recommendation "日常低强度散步 20-30 分钟(心率 < 100 bpm, 轻松交谈), 或练习太极 20 分钟。不占用系统糖原与神经恢复储备。"
  }
}

day "D8"
name "主动休整日 B"
type "rest"
duration "自由安排"
recommendation "日常户外散步 20-30 分钟, 接触自然光; 保障夜间深睡, 准备进入下一轮 D1。"
}
}

```

三、前端核心系统实现代码

以下为基于现代前端标准 (Vue 3 Composition API + Tailwind CSS + Lucide Icons) 编写的完整界面与逻辑实现。其内嵌了红黄绿灯三色卡片机制、滑动手势备忘抽屉 (包含器械替代、Linda 死手守则与防衰指标)、爆发力向心辅助提示器与热身平衡打卡。

```

HTML
<!-- index.html / WorkoutV6App.vue -->
<template>
  <div class="min-h-screen bg-slate-950 text-slate-100 flex flex-col font-sans pb-28 select-none">

    <!-- 顶部状态栏与用户切换 -->
    <header class="sticky top-0 z-40 bg-slate-900/90 backdrop-blur-md border-b border-slate-800 px-4 py-3">
      <div class="max-w-2xl mx-auto flex items-center justify-between">
        <div class="flex items-center space-x-2">
          <div class="w-8 h-8 rounded-lg bg-emerald-500/20 border border-emerald-500/40 flex items-center justify-center font-black text-emerald-400 text-sm">
            V6
          </div>
          <div>
            <h1 class="text-sm font-bold text-white tracking-wide"> 天循环计划: 防衰双目标 </h1>
            <p class="text-[11px] text-slate-400"> 耐力 | 心肺 | 爆发力 | 骨密度平衡 </p>
          </div>
        </div>
      </div>

      <div class="flex items-center space-x-2">

```

```

<!-- 切换 Leo / Linda -->
<div class="bg-slate-800 p-0.5 rounded-lg flex border border-slate-700 text-xs">
  <button
    @click="currentUser = 'Leo'"
    :class="currentUser === 'Leo' ? 'bg-indigo-600 text-white font-semibold' : 'text-slate-400'"
    class="px-2.5 py-1 rounded-md transition-all">
    Leo
  </button>
  <button
    @click="currentUser = 'Linda'"
    :class="currentUser === 'Linda' ? 'bg-rose-600 text-white font-semibold' : 'text-slate-400'"
    class="px-2.5 py-1 rounded-md transition-all">
    Linda
  </button>
</div>

```

```

<!-- 打开备忘卡库按钮 -->
<button
  @click="isMemoOpen = true"
  class="px-2.5 py-1 rounded-lg bg-amber-500/15 border border-amber-500/40
text-amber-300 text-xs font-semibold flex items-center space-x-1 hover:bg-amber-500/25
active:scale-95 transition">
  <span>📅</span>
  <span>备忘</span>
</button>
</div>
</div>
</header>

```

```

<!-- 主容器 -->
<main class="max-w-2xl mx-auto w-full px-4 pt-4 flex-1">

```

```

<!-- 8天循环滑动切换指示条 -->
<section class="mb-4">
  <div class="flex items-center justify-between text-xs text-slate-400 mb-2">
    <span>📅 大概周期进度</span>
    <span class="text-emerald-400 font-mono">{{ (activeDayIndex + 1) / Dk }}</span>
  </div>
  <div class="grid grid-cols-8 gap-1.5 bg-slate-900/60 p-1.5 rounded-xl border
border-slate-800/80">
    <button
      v-for="(day, idx) in daysList"
      :key="day.day"
      @click="activeDayIndex = idx"

```

```

      :class="[
        activeDayIndex === idx ? 'bg-indigo-600 text-white shadow-lg shadow-indigo-500/30
font-bold' : 'bg-slate-800/50 text-slate-400 hover:bg-slate-800',
        day.type === 'rest' ? 'border-dashed border border-slate-700' : ""
      ]"
      class="py-2 rounded-lg text-xs flex flex-col items-center justify-center transition-all">
      <span class="text-[10px] opacity-75"> {day} {day} </span>
      <span class="text-[11px] mt-0.5">
        {
          (day.type === 'strength' ? 力 : (day.type === 'hlt' ? 4×4 : (day.type === 'cardio' ?
Z2 : 休)))
        }
      </span>
    </button>
  </div>
</section>

```

```

<!-- 需求 (1): 绿灯 / 黄灯 / 红灯状态选择卡片 -->
<section class="mb-5 bg-slate-900 border border-slate-800 rounded-2xl p-4 shadow-xl">
  <div class="flex items-center justify-between mb-3">
    <div class="flex items-center space-x-1.5">
      <span class="text-xs font-bold uppercase tracking-wider text-slate-400"> 今日身体就绪状
态: </span>
      <span class="text-[11px] text-slate-500 font-mono"> 决定训练强度与模式 </span>
    </div>
    <span>
      :class="{
        'text-emerald-400 bg-emerald-500/10 border-emerald-500/30': statusLight === 'green',
        'text-amber-400 bg-amber-500/10 border-amber-500/30': statusLight === 'yellow',
        'text-rose-400 bg-rose-500/10 border-rose-500/30': statusLight === 'red'
      }"
      class="text-[11px] px-2 py-0.5 rounded-full border font-semibold">
        {
          (statusLight === 'green' ? 绿灯 : 满血输出 : (statusLight === 'yellow' ? 黄灯 : 保守强
制 : 红灯 : 防衰储备))
        }
      </span>
    </div>

```

```

<!-- 三灯选择按钮 -->
<div class="grid grid-cols-3 gap-2.5">
  <!-- 绿灯 -->
  <button
    @click="statusLight = 'green'"
    :class="statusLight === 'green' ? 'ring-2 ring-emerald-400 bg-emerald-950/40
border-emerald-500/50' : 'bg-slate-800/60 border-slate-700/60 opacity-60'"
    class="p-2.5 rounded-xl border text-left transition-all relative overflow-hidden flex flex-col
justify-between">

```

```
<div class="flex items-center space-x-1.5 mb-1">
  <span class="w-3 h-3 rounded-full bg-emerald-400 shadow-sm
shadow-emerald-400"></span>
  <span class="text-xs font-bold text-emerald-200">黄灯 (A方案) </span>
</div>
<p class="text-[10px] text-slate-300 leading-tight">自由=器械等 <br>激活盲项爆发同
</p>
</button>
```

```
<!-- 黄灯 -->
<button
@click="statusLight = 'yellow'"
:class="statusLight === 'yellow' ? 'ring-2 ring-amber-400 bg-amber-950/40
border-amber-500/50' : 'bg-slate-800/60 border-slate-700/60 opacity-60'"
class="p-2.5 rounded-xl border text-left transition-all relative overflow-hidden flex flex-col
justify-between">
  <div class="flex items-center space-x-1.5 mb-1">
    <span class="w-3 h-3 rounded-full bg-amber-400 shadow-sm
shadow-amber-400"></span>
    <span class="text-xs font-bold text-amber-200">黄灯 (B方案) </span>
  </div>
  <p class="text-[10px] text-slate-300 leading-tight">自由=器械等 <br>激活盲项爆发同
</p>
</button>
```

```
<!-- 红灯 -->
<button
@click="statusLight = 'red'"
:class="statusLight === 'red' ? 'ring-2 ring-rose-400 bg-rose-950/40 border-rose-500/50' :
'bg-slate-800/60 border-slate-700/60 opacity-60'"
class="p-2.5 rounded-xl border text-left transition-all relative overflow-hidden flex flex-col
justify-between">
  <div class="flex items-center space-x-1.5 mb-1">
    <span class="w-3 h-3 rounded-full bg-rose-400 shadow-sm shadow-rose-400"></span>
    <span class="text-xs font-bold text-rose-200">红灯 (休整) </span>
  </div>
  <p class="text-[10px] text-slate-300 leading-tight">切换主动恢复 <br>10-30分散步/拉伸
</p>
</button>
</div>
```

```
<!-- 灯态执行说明条文 -->
<div class="mt-3 pt-2.5 border-t border-slate-800 text-[11px] text-slate-400 flex items-start
space-x-1.5">
```

```
<span class="text-slate-500">💡 </span>
<span v-if="statusLight === 'green'">
    当前状态良好，力量主动作开放自由版切换。正式组前2次采用最大意图向心爆发（~1s）
    第3次恢复高跳速度。
</span>
<span v-else-if="statusLight === 'yellow'" class="text-amber-300/90">
    疲劳警戒，系统已自动屏蔽自由重量切换，切回纯器械稳定轨道，取消爆发意图，全指
    控制离心，保动作质量。
</span>
<span v-else class="text-rose-300/90">
    伤痛/极度疲劳阻断，严禁执行高负荷抗阻。主课表已替换为低心率主动恢复散步、前
    肢放松与关节活动度。
</span>
</div>
</section>
```

```
<!-- 红灯状态下的替换视图：防衰主动恢复卡 -->
<section v-if="statusLight === 'red'" class="bg-rose-950/20 border border-rose-500/30
rounded-2xl p-5 mb-5">
    <div class="flex items-center space-x-2 text-rose-400 mb-3">
        <span class="text-xl">🔴 </span>
        <h2 class="text-base font-bold">红灯保护机制已生效，今日转为主动作版 </h2>
    </div>
    <p class="text-xs text-slate-300 leading-relaxed mb-4">
        ACSM 老年健康指引：在急性疼痛、发热、严重睡眠缺失或眩晕状态下进行大负荷抗阻
        神经对稳定肌的支配能力骤降，跌倒风险成倍增加。今日任务是促进血液回流与组织修复。
    </p>
```

```
<div class="space-y-3">
    <div class="p-3 bg-slate-900/80 rounded-xl border border-rose-900/40 flex items-center
justify-between">
        <div>
            <div class="text-xs font-bold text-white">⚠️ 避免户外/跑步机散步 </div>
            <div class="text-[11px] text-slate-400">0-30 分钟，心率控制在 100 bpm 以下，能轻松
交谈 </div>
        </div>
        <input type="checkbox" v-model="redDayRecord.walkDone" class="w-5 h-5 rounded
text-rose-600 focus:ring-0">
    </div>
```

```
<div class="p-3 bg-slate-900/80 rounded-xl border border-rose-900/40 flex items-center
justify-between">
    <div>
```

```

        <div class="text-xs font-bold text-white">温和全身拉伸与泡沫轴放松</div>
        <div class="text-[11px] text-slate-400">10-15 分钟，重点小腿腓肠肌、髂胫肌、胸大肌
    </div>
    </div>
    <div>
        <input type="checkbox" v-model="redDayRecord.stretchDone" class="w-5 h-5 rounded
text-rose-600 focus:ring-0">
    </div>
    </div>

    <button
        @click="logActiveRecovery"
        class="mt-5 w-full py-2.5 rounded-xl bg-rose-600 hover:bg-rose-500 font-bold text-xs
text-white shadow-lg shadow-rose-900/40">
        打卡保存今日主动恢复日志
    </button>
</section>

<!-- 绿灯/黄灯状态下的常规课表展示 -->
<section v-else>
    <!-- 当日概览头 -->
    <div class="flex items-center justify-between mb-3">
        <div>
            <span class="text-xs font-mono font-bold text-indigo-400 uppercase tracking-wider">
currentDayPlan.day</span>
            <h2 class="text-base font-black text-white">{{currentDayPlan.name}}</h2>
        </div>
        <div class="text-right">
            <span class="text-xs text-slate-400">当前课时</span>
            <div class="text-xs font-mono font-bold text-emerald-400">{{currentDayPlan.duration}}
        </div>
    </div>
    </div>
    </div>

    <!-- 需求 (3) 架构调整: 热身模块内嵌5分钟平衡与步态打卡 (力量日) -->
    <div v-if="currentDayPlan.warmup && currentDayPlan.warmup.balanceMin > 0" class="mb-4
bg-slate-900/90 border border-indigo-900/30 rounded-2xl p-4">
        <div class="flex items-center justify-between mb-2">
            <div class="flex items-center space-x-2">
                <span class="text-indigo-400 font-bold text-xs">🎯 完整热身 (12分钟)</span>
                <span class="text-[10px] bg-indigo-500/20 text-indigo-300 px-1.5 py-0.5 rounded">后5分钟
平衡防跌倒</span>
            </div>
            <label class="flex items-center space-x-1.5 text-xs text-slate-300">

```

```

      <input type="checkbox" v-model="dayWarmupChecked" class="w-4 h-4 rounded
text-indigo-600">
    </span> 完成热身 </span>
  </label>
</div>

```

```

    <ul class="text-[11px] text-slate-400 space-y-1 pl-1 list-disc list-inside">
      <li v-for="(bItem, idx) in currentDayPlan.warmup.balanceItems" :key="idx"
class="text-slate-300">
        {{ bItem }}
      </li>
    </ul>
  </div>

```

```

<!-- 力量日动作清单 -->
<div v-if="currentDayPlan.type === 'strength'" class="space-y-4">
  <div
    v-for="(exercise, exIdx) in currentDayPlan.exercises"
    :key="exercise.id"
    class="bg-slate-900 border border-slate-800 rounded-2xl p-4 shadow-md transition-all">

```

```

    <div class="flex items-start justify-between">
      <div>
        <div class="flex items-center space-x-2">
          <span class="text-xs font-mono text-slate-500">#0{{ exIdx + 1 }} </span>
          <h3 class="text-sm font-bold text-white">
            {{ (selectedEquip[exercise.id] === 'free' && statusLight === 'green') ?
exercise.freeEquip : exercise.name }}
          </h3>
        </div>

```

```

<!-- 需求 (3): 首复合动作爆发意图向心提示 (仅绿灯生效) -->
<div v-if="exercise.isFirstCompound" class="mt-1 flex items-center space-x-1.5">
  <span
    :class="statusLight === 'green' ? 'bg-amber-500/20 text-amber-300
border-amber-500/40' : 'bg-slate-800 text-slate-500 border-slate-700'"
    class="text-[10px] font-bold px-1.5 py-0.2 rounded border">
    {{ statusLight === 'green' ? ⚡ 爆发向心 前2次最快速度(1s) 匀速向心 (黄灯出
过爆发) }}
  </span>
</div>
</div>

```

```

<!-- 需求 (3): 自由重量 vs 器械 等效切换按钮 -->

```

```

<div v-if="exercise.freeEquiv && statusLight === 'green'" class="flex items-center space-x-1
bg-slate-800 p-0.5 rounded-lg border border-slate-700 text-[10px]">
  <button
    @click="selectedEquiv[exercise.id] = 'machine'"
    :class="selectedEquiv[exercise.id] !== 'free' ? 'bg-slate-700 text-white font-bold' :
'text-slate-400'"
    class="px-2 py-0.5 rounded">
    器械
  </button>
  <button
    @click="toggleFreeEquiv(exercise)"
    :disabled="currentUser === 'Linda' && !exercise.lindaAllowedFree"
    :class="[
      selectedEquiv[exercise.id] === 'free' ? 'bg-indigo-600 text-white font-bold' :
'text-slate-400',
      (currentUser === 'Linda' && !exercise.lindaAllowedFree) ? 'opacity-30 cursor-not-allowed'
      : ''
    ]"
    class="px-2 py-0.5 rounded flex items-center space-x-1">
    <span>自由版</span>
    <span v-if="currentUser === 'Linda' && !exercise.lindaAllowedFree"
class="text-[9px]">🚫</span>
  </button>
</div>
</div>

```

```

<!-- Linda 死手模式合规提醒 -->
<div v-if="currentUser === 'Linda' && exercise.id === 'd4-ex1'" class="mt-2 text-[11px]
bg-rose-500/10 text-rose-300 border border-rose-500/20 rounded-lg p-2 flex items-center
space-x-1.5">
  <span>🔒</span>
  <span><strong> Linda 死手合规提醒 </strong> 已纳入 RDL，请务必使用金属万向助力钩连接
接杠铃/六角杠</span>
</div>

```

```

<!-- 组数、次数、RPE -->
<div class="grid grid-cols-4 gap-2 mt-3 pt-3 border-t border-slate-800/80 text-center
font-mono">
  <div class="bg-slate-950/60 p-1.5 rounded-lg">
    <span class="text-[10px] text-slate-500 block">E 式组</span>
    <span class="text-xs font-bold text-white"> (statusLight === 'yellow' ? Math.max(2,
exercise.sets - 1) : exercise.sets) 组</span>
  </div>
  <div class="bg-slate-950/60 p-1.5 rounded-lg">

```

```

    <span class="text-[10px] text-slate-500 block">目标次数</span>
    <span class="text-xs font-bold text-white"> exercise.reps </span>
  </div>
  <div class="bg-slate-950/60 p-1.5 rounded-lg">
    <span class="text-[10px] text-slate-500 block">RPE 目标</span>
    <span class="text-xs font-bold text-emerald-400"> statusLight === 'yellow' ? '7-8' :
exercise.rpe </span>
  </div>
  <div class="bg-slate-950/60 p-1.5 rounded-lg">
    <span class="text-[10px] text-slate-500 block">组间间歇</span>
    <span class="text-xs font-bold text-indigo-400"> exercise.rest </span>
  </div>
</div>

```

```

    <p class="mt-2 text-[11px] text-slate-400 leading-normal"> { exercise.notes } </p>
  </div>
</div>

```

```

<!-- 心肺日 / HIIT 日 / 休息日专用卡片 -->
<div v-else-if="currentDayPlan.type === 'cardio'" class="bg-slate-900 border border-slate-800
rounded-2xl p-5 space-y-4">
  <div class="flex items-center justify-between">
    <h3 class="text-sm font-bold text-emerald-400"> 🏃 D3 Zone 2 坡度快走执行规程 </h3>
    <span class="text-xs bg-emerald-500/20 text-emerald-300 px-2 py-0.5 rounded-full">
坡度+心肺</span>
  </div>
  <div class="p-3 bg-slate-950 rounded-xl space-y-1.5 text-xs text-slate-300 border
border-slate-800">
    <div><strong>坡度指标</strong> { currentDayPlan.cardioProtocol.target } </div>
    <div><strong>心率指标</strong> { currentDayPlan.cardioProtocol.main } </div>
    <div class="text-[11px] text-slate-400"> 注：椭圆机为下肢减负设计，骨密度刺激不如坡
度快走 </div>
  </div>
  <div class="text-xs text-slate-400">
    <strong>心肺与拉伸 (15分) </strong> { currentDayPlan.cardioProtocol.core }，随后进
行 10 分钟系统柔韧拉伸
  </div>
</div>

```

```

<div v-else-if="currentDayPlan.type === 'hiit'" class="bg-slate-900 border border-slate-800
rounded-2xl p-5 space-y-4">
  <div class="flex items-center justify-between">
    <h3 class="text-sm font-bold text-rose-400"> 🔥 D6 挪威 4×4 间歇跑规程 </h3>

```

```

    <span class="text-xs bg-rose-500/20 text-rose-300 px-2 py-0.5 rounded-full">VO2 Max 金
  </span>
</div>
</div>
    <div class="p-3 bg-slate-950 rounded-xl space-y-2 text-xs text-slate-300 border
border-slate-800">
      <div><strong> 阶段 I </strong> 85-95% HRmax, 耐力突破最大摄氧量 </div>
      <div><strong> 阶段 II </strong> 4 轮 (4分钟高速跑 + 3分钟低速活动恢复) </div>
      <div class="text-rose-400 text-[11px]">⚠️ 心率限制: 训练结束后到下一轮 D1 必须满 2
小时恢复! </div>
    </div>
  </div>
</div>

```

```

    <div v-else-if="currentDayPlan.type === 'rest'" class="bg-slate-900 border border-slate-800
rounded-2xl p-5 space-y-4">
      <h3 class="text-sm font-bold text-indigo-400">🛌 休息日主动活动指南 </h3>
      <p class="text-xs text-slate-300 leading-relaxed">currentDayPlan.recommendation
    </p>
      <div class="p-3 bg-slate-950 rounded-xl border border-slate-800 flex items-center
justify-between">
        <div>
          <div class="text-xs font-bold text-white">记录今日轻松散步 (20-30分) </div>
          <div class="text-[11px] text-slate-500">心率 < 100 bpm · 维持每周 4 次以上有氧刺激
        </div>
      </div>
      <div>
        <input type="checkbox" v-model="restDayWalkChecked" class="w-5 h-5 rounded
text-indigo-600">
      </div>
    </div>
  </div>
</section>

```

```

<!-- 需求 (3) 架构调整: 底部常驻功能性指标快速录入 (握力 & 单腿站立防衰指标) -->
<section class="mt-6 bg-slate-900/60 border border-slate-800 rounded-2xl p-4">
  <div class="flex items-center justify-between mb-3">
    <span class="text-xs font-bold text-slate-300 flex items-center space-x-1">
      <span>📊</span>
      <span>功能性防衰指标追踪 (V6 新增)</span>
    </span>
    <span class="text-[10px] text-slate-500">定期测试 · 评估肌力与神经募集 </span>
  </div>

```

```

  <div class="grid grid-cols-2 gap-3">
    <div class="bg-slate-950 p-2.5 rounded-xl border border-slate-800 flex items-center
justify-between">

```

```

    <div>
      <span class="text-[10px] text-slate-500 block"> 握力年度基线 (kg) </span>
      <span class="text-[10px] text-slate-400"> 男 < 16 / 女 < 16 低下 </span>
    </div>
    <input
      type="number"
      placeholder="测试值"
      v-model="functionalMetrics.grip"
      class="w-16 bg-slate-900 border border-slate-700 rounded-lg px-2 py-1 text-xs text-right
font-mono text-emerald-400">
    </div>

```

```

    <div class="bg-slate-950 p-2.5 rounded-xl border border-slate-800 flex items-center
justify-between">
      <div>
        <span class="text-[10px] text-slate-500 block"> 单腿睁眼站立 (秒) </span>
        <span class="text-[10px] text-slate-400"> 与跌倒目标 > 30秒 </span>
      </div>
      <input
        type="number"
        placeholder="秒数"
        v-model="functionalMetrics.balanceSec"
        class="w-16 bg-slate-900 border border-slate-700 rounded-lg px-2 py-1 text-xs text-right
font-mono text-indigo-400">
      </div>
    </div>
  </section>

```

```

</main>

```

```

<!-- 需求 (2): 备忘卡区 (点开后左右滑动查找备忘卡片) -->

```

```

<div
  v-if="isMemoOpen"
  class="fixed inset-0 z-50 bg-black/80 backdrop-blur-sm flex flex-col justify-end
transition-opacity">

```

```

  <!-- 遮罩点击关闭 -->
  <div class="flex-1" @click="isMemoOpen = false"></div>

```

```

  <!-- 滑动卡片容器 -->
  <div class="bg-slate-900 border-t border-slate-800 rounded-t-3xl p-5 max-w-2xl mx-auto
w-full shadow-2xl flex flex-col max-h-[85vh]">

```

```

  <!-- 抽屉把手和标题 -->

```

```

<div class="flex items-center justify-between pb-3 border-b border-slate-800">
  <div class="flex items-center space-x-2">
    <span class="text-base"> 📝 </span>
    <h3 class="text-sm font-bold text-white"> 备忘录管理 </h3>
  </div>
  <button
    @click="isMemoOpen = false"
    class="text-xs bg-slate-800 text-slate-400 hover:text-white px-2.5 py-1 rounded-full border
border-slate-700">
    关闭
  </button>
</div>

```

```

<p class="text-[11px] text-slate-400 mt-2 mb-3">
  👉 左右滑动卡片，可查看卡片英文原稿、Linda双手模式、盲密度阶梯与测试备注
</p>

```

```

<!-- 横向滑动容器 (Scroll-Snap Slider) -->
<div class="flex space-x-4 overflow-x-auto pb-4 pt-1 snap-x snap-mandatory scrollbar-none
flex-1">

```

```

  <div
    v-for="memo in memoCards"
    :key="memo.id"
    class="min-w-[85%] sm:min-w-[75%] snap-center bg-slate-950 border border-slate-800
rounded-2xl p-4.5 flex flex-col justify-between shadow-lg">

```

```

    <div>
      <div class="flex items-center justify-between mb-2">
        <span
          :class="[
            memo.color === 'blue' ? 'bg-blue-500/20 text-blue-300 border-blue-500/30' : '',
            memo.color === 'rose' ? 'bg-rose-500/20 text-rose-300 border-rose-500/30' : '',
            memo.color === 'amber' ? 'bg-amber-500/20 text-amber-300 border-amber-500/30' : '',
            memo.color === 'emerald' ? 'bg-emerald-500/20 text-emerald-300
border-emerald-500/30' : '',
            memo.color === 'purple' ? 'bg-purple-500/20 text-purple-300 border-purple-500/30' : '',
          ]"
          class="text-[10px] font-bold px-2 py-0.5 rounded-full border">
            {{ memo.tag }}
          </span>
          <span v-if="memo.badge" class="text-[10px] text-rose-400 font-mono"> {{ memo.badge }}
        </span>
      </div>
    </div>

```

```
<h4 class="text-sm font-bold text-white mb-3">{{ memo.title }}</h4>
```

```
<!-- 器械等效表格形式渲染 -->
```

```
<div v-if="memo.id === 'memo-swap'" class="space-y-2 text-xs">
```

```
<div class="overflow-x-auto">
```

```
<table class="w-full text-left text-[11px] border-collapse">
```

```
<thead>
```

```
<tr class="text-slate-500 border-b border-slate-800">
```

```
<th class="py-1">日期</th>
```

```
<th class="py-1">器械名称</th>
```

```
<th class="py-1">器械等效性</th>
```

```
</tr>
```

```
</thead>
```

```
<tbody class="divide-y divide-slate-800/60 text-slate-300 font-mono text-[11px]">
```

```
<tr v-for="(item, i) in memo.content" :key="i">
```

```
<td class="py-1.5 text-indigo-400">{{ item.day }}</td>
```

```
<td class="py-1.5">{{ item.machine }}</td>
```

```
<td class="py-1.5 text-emerald-300">{{ item.free }}</td>
```

```
</tr>
```

```
</tbody>
```

```
</table>
```

```
</div>
```

```
<div class="mt-3 space-y-1 text-[11px] text-slate-400 pt-2 border-t border-slate-800">
```

```
<p v-for="(r, ri) in memo.rules" :key="ri" class="leading-tight">{{ r }}</p>
```

```
</div>
```

```
</div>
```

```
<!-- 其余清单式备忘卡渲染 -->
```

```
<div v-else class="space-y-2.5 text-xs text-slate-300">
```

```
<div
```

```
  v-for="(line, li) in memo.content"
```

```
  :key="li"
```

```
  class="p-2.5 rounded-xl bg-slate-900/70 border border-slate-800/60 leading-relaxed text-[11px]">
```

```
    {{ line }}
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<div class="text-[10px] text-slate-500 text-right mt-3">
```

```
  V6 防衰规范卡 · 对应 2026.10 周地
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</template>
```

```
<script setup>
```

```
import { ref, reactive, computed } from 'vue';
```

```
import { V6_DAYS_SCHEDULE, MEMO_CARDS } from './v6-workout-data.js';
```

```
// 当前用户切换
```

```
const currentUser = ref('Leo'); // 'Leo' | 'Linda'
```

```
// 需求 (1): 绿灯 / 黄灯 / 红灯状态响应式变量
```

```
const statusLight = ref('green'); // 'green' | 'yellow' | 'red'
```

```
// 需求 (2): 备忘抽屉开关与卡片数据
```

```
const isMemoOpen = ref(false);
```

```
const memoCards = ref(MEMO_CARDS);
```

```
// 课表状态
```

```
const daysList = ref(V6_DAYS_SCHEDULE);
```

```
const activeDayIndex = ref(0);
```

```
const currentDayPlan = computed(() => daysList.value[activeDayIndex.value]);
```

```
// 自由动作等效切换状态存储 (默认器械版)
```

```
const selectedEquiv = reactive({});
```

```
// 热身与记录项打卡状态
```

```
const dayWarmupChecked = ref(false);
```

```
const restDayWalkChecked = ref(false);
```

```
const redDayRecord = reactive({
```

```
  walkDone: false,
```

```
  stretchDone: false
```

```
});
```

```
// 需求 (3): V6 新增功能性指标存储
```

```
const functionalMetrics = reactive({
```

```
  grip: "",
```

```
  balanceSec: "
```

```
});
```

```
// 自由版切换约束处理(Linda死手拦截)
function toggleFreeEquiv(exercise) {
  if (currentUser.value === 'Linda' && !exercise.lindaAllowedFree) {
    alert("【Linda 死手模式拦截】: 该动作涉及小臂代偿, Linda严禁使用自由版本!");
    return;
  }
  selectedEquiv[exercise.id] = selectedEquiv[exercise.id] === 'free' ? 'machine' : 'free';
}

function logActiveRecovery() {
  alert("今日红灯主动恢复已记录! 系统已保全恢复周期。");
}
</script>

<style scoped>
/* 隐藏原生横向滚动条, 保证滑动卡片如 App 般丝滑 */
.scrollbar-none::-webkit-scrollbar {
  display: none;
}
.scrollbar-none {
  -ms-overflow-style: none;
  scrollbar-width: none;
}
</style>
```

四、本地部署与项目整改落地步骤

1. 替换/新增核心课表模块：
 - 将上述配置保存到项目的 src/data/v6-workout-data.js 中。
 - 如果项目为静态单文件(例如基于 CDN Vue3 + Tailwind 的 index.html), 可直接把 V6_PLAN_CONFIG、MEMO_CARDS 及对应模版注入原有的主 div#app 中。
2. 清除旧版遗留的硬编码判定：
 - 搜索并移除旧版代码中的 RDL is optional、B-Stance default only 逻辑, 替换为当前的绿灯日平级并列选择逻辑。
 - 搜索 Linda 动作过滤代码, 更新为「允许使用金属宽面钩执行杠铃与六角杠 RDL」。
3. 测试三种核心场景的交互：
 - 测试绿灯: 切换至 D1 或 D4, 检查第一项动作是否出现 ⚡ 爆发向心 徽标; 点击「自由版」, 哈克深蹲切换为史密斯深蹲/六角杠硬拉, B-Stance 切换为杠铃/六角杠 RDL。
 - 测试黄灯: 点击「黄灯」, 系统应立即将自由版按钮禁用并折叠, 提示隐藏爆发向心, 组数提示减 1, RPE 显示 7-8。
 - 测试红灯: 点击「红灯」, 当前主课表隐藏, 出现 20-30 分钟低心率散步与拉伸的主动恢复卡。
 - 测试备忘卡区: 点击右上角 📝 备忘智库, 左右横向滑动手感需流畅, 检查 Linda 死手模式、DXA 骨密度、跳跃进阶等卡片内容是否完整呈现。
4. 重新构建与 Docker 部署：

执行工程目录下的构建命令：

```
Bash
```

```
npm run build
```

```
# 或直接运行原部署脚本
```

```
bash deploy.sh
```

部署至 NAS 上的 Nginx 容器后，即可在手机移动端上体验顺滑的滑动备忘卡与红绿灯自适应模式。